Explainable Machine Learning and Grounded Policy Retrieval Framework for Food Inflation Forecasting and Economic Early Warning in Bangladesh

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Keywords— *Food Inflation Nowcasting, Explainable Machine Learning, Economic Early Warning, Conformal Prediction, Policy Information Retrieval.*

1. Research Area

Applied Machine Learning, Computational Macroeconomics, Empirical Time-Series Nowcasting, and Information Retrieval.

2. Goals/Objectives

This research investigates macroeconomic volatility and staple food price forecasting in developing, import-dependent economies, focusing empirically on Bangladesh. In official surveillance, Consumer Price Index (CPI) and foreign exchange reserve releases from the Bangladesh Bureau of Statistics (BBS) and Bangladesh Bank typically appear around Day 7 to Day 10 of following month (Month M+1). While this represents a modest administrative delay of one to two weeks, rapid international commodity and currency fluctuations during shock regimes create an informational gap for monetary and procurement authorities. The core objective is to evaluate whether non-linear machine learning architectures can extract actionable early signals from high-frequency international markets and domestic macro indicators ahead of official monthly bulletins. Specifically, the study investigates: (i) whether gradient boosting improves short-horizon food inflation nowcasting relative to linear regression and naive persistence baselines; (ii) how empirical conformal prediction behaves when calibrating retail food price uncertainty bands; (iii) whether unsupervised anomaly detection can provide advance warning before systemic macroeconomic distress episodes; and (iv) how policy retrieval pipelines perform when grounding domain economic inquiries on central bank publications.

3. Methodology

The empirical architecture is anchored on a continuous daily macroeconomic panel spanning January 2010 to September 2026 (6,117 daily observations, 41 indicators) combined with an official monthly macroeconomic panel (201 continuous months, zero synthetic interpolation). Institutional release schedules (BBS CPI on Day 10 of M+1, Bangladesh Bank reserves on Day 7 of M+1, WFP retail food price surveys on Day 15 of M) are strictly enforced via step-function forward fills to prevent look-ahead bias, while daily international shocks (Brent crude oil, CBOT wheat futures, US Dollar Index, natural gas, gold) enter contemporaneously. Annual structural indicators from the World Bank (external debt, import coverage ratios) are incorporated on an as-of release basis, though annual step-filling introduces an open look-ahead timing consideration. For food inflation nowcasting, an Extreme Gradient Boosting (XGBoost) model is trained using chronological walk-forward splits and compared against linear Ridge regression, Random Forest, and naive persistence baselines using Harvey-Leybourne-Newbold corrected Diebold-Mariano tests. Feature attributions are extracted via TreeSHAP. For staple retail commodities (coarse rice, lentils, wheat flour), a rolling walk-forward adaptive conformal prediction engine is evaluated over a 730-day calibration window to generate distribution-free 80% nominal coverage intervals. For crisis surveillance, an Isolation Forest is trained exclusively on pre-2022 historical controls and evaluated out-of-sample on 2022–2026 stress regimes with a reset-capable Cumulative Sum (CUSUM) filter. A prototype dense retrieval module (ChromaDB) indexes authentic central bank policy texts.

4. Results and Analysis

In monthly out-of-sample evaluation (2024–2026, $n \approx 33$ months), XGBoost reduces root mean square error (RMSE) relative to a linear Ridge baseline by 20.0% at the t+1 horizon (1.171% vs. 1.464%) and by 32.5% at the t+3 horizon (1.740% vs. 2.578%), but neither difference is statistically significant at the 5% level (HLN-corrected Diebold-Mariano $p = 0.068$, 0.058), and XGBoost does not outperform naive persistence (1.160% at t+1) or Random Forest (1.087% at t+1, 1.725% at t+3) as detailed in Table I. In high-frequency daily 30-day trajectory forecasting, XGBoost substantially outperforms Ridge (RMSE 1.158% vs. 2.457%, a 52.9% reduction; DM = 3.74, $p < 0.001$), yet remains marginally behind naive persistence (1.137%), indicating dominant short-term price stickiness in domestic staple markets. TreeSHAP feature attribution assigns the highest importance to reserve-adequacy indicators (months of imports: 19.3%), remittance inflows (15.9%), and headline inflation persistence (15.4%), while global crude oil (1.4%) and CBOT wheat (1.2%) exhibit minor direct attribution; annual structural indicators among them remain an open look-ahead concern.

Forecast Horizon	Naive Baseline	Linear Ridge	Random Forest	Proposed XGBoost	DM Stat vs Ridge (p-val)
t+1 (1 Month Ahead)	1.160%	1.464%	1.087%	1.171% (-20.0%)	1.82 (p = 0.068)
t+3 (3 Months Ahead)	1.757%	2.578%	1.725%	1.740% (-32.5%)	1.90 (p = 0.058)
30-Day Forward Daily	1.137%	2.457%	—	**1.158% (-52.9%)**	**3.74 (p < 0.001)**

Table I: Out-of-Sample Forecasting Performance & Diebold-Mariano Tests across Horizons

In retail price uncertainty estimation, rolling walk-forward conformal prediction achieved 85.1% empirical out-of-sample coverage for coarse rice over 1,366 unseen test days, whereas lentils (74.8%) and flour (75.1%) fell short of the 80% nominal target, highlighting commodity-specific calibration disparities. The unsupervised anomaly detector flagged both out-of-sample economic crisis episodes 80 days (2022 foreign exchange drain) and 90 days (2024 political shock) ahead of onset, but flagged 74.0% of out-of-sample days due to persistent post-2022 macroeconomic disequilibrium, indicating that raw alert frequencies must be coupled with secondary persistence filters before operational deployment. The retrieval module was audited for keyword and semantic coverage, achieving an 82.0% mean grounding score across 10 central bank queries; full LLM generation and hallucination benchmarks have not yet been evaluated. A prototype web dashboard provides interactive scenario exploration.

5. References

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